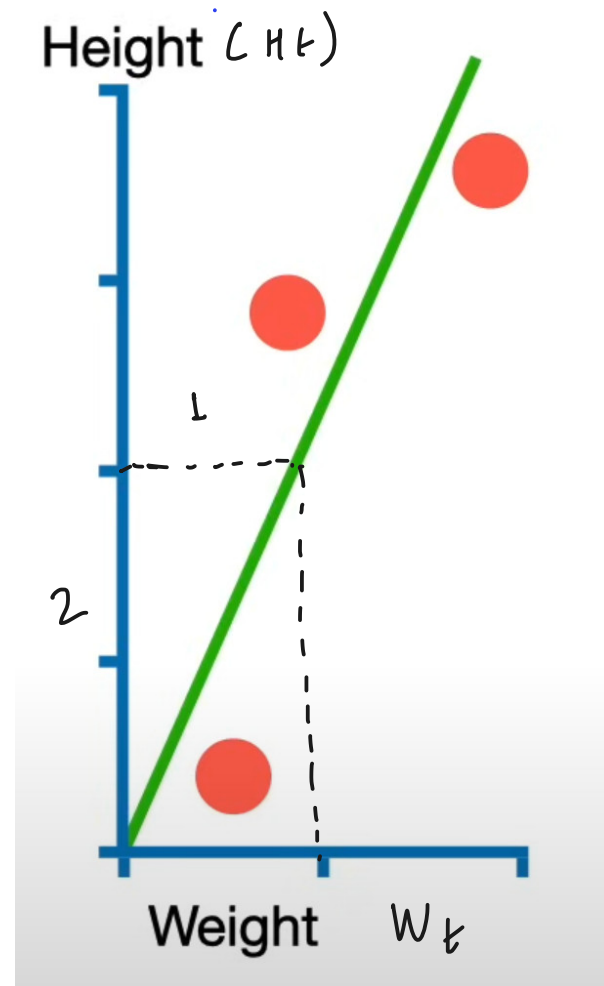


Chain Rule

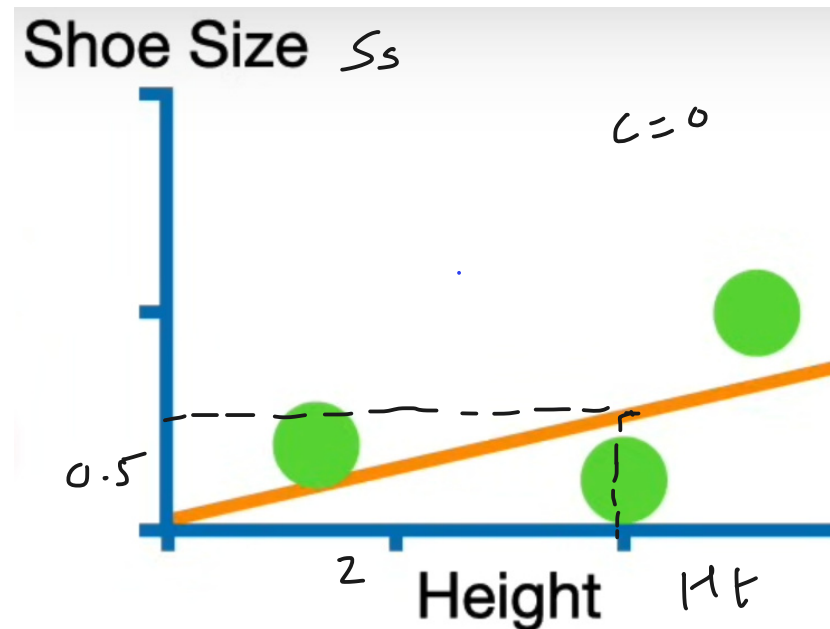
Predict shoe size (S_s) from weight (W_t)



$$y = mx + c$$

$$m = \frac{dH_t}{dW_t} \quad c = 0$$

$$H_t = \frac{dH_t}{dW} W_t \quad \text{--- (1)}$$



$$y = mx + c$$

$$m = \frac{dS_s}{dH_t} \quad c = 0$$

$$S_s = \frac{dS_s}{dH_t} H_t \quad \text{--- (2)}$$

using eq (1) substitute H_t

$$S_s = \frac{dS_s}{dH_t} \frac{dH_t}{dW_t} W_t \quad \text{--- (3)}$$

- To calculate shoe size directly from weight we can write

$$y = mx + c$$

$$m = \frac{dS_s}{dW_t} \quad c = 0$$

$$S_s = \frac{dS_s}{dW_t} W_t \quad \text{--- (4)}$$

From eq (3) & eq (4) we can write

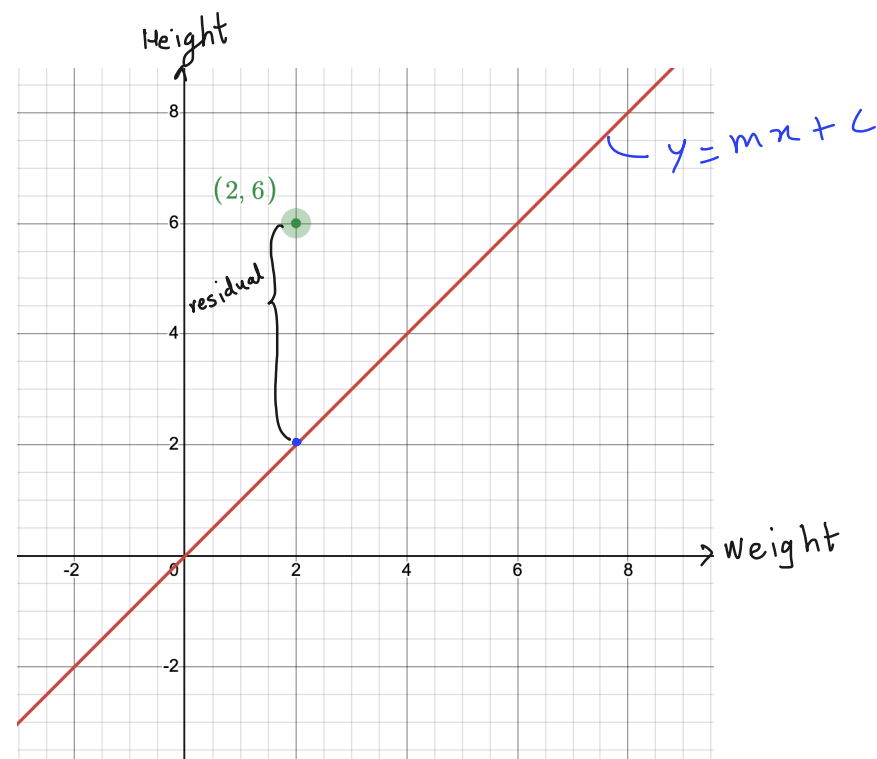
Chain Rule

$$\frac{dS_s}{dW_t} = \frac{dS_s}{dH_t} \times \frac{dH_t}{dW_t}$$

Chain Rule

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

Chain Rule in Regression



eqⁿ of line $y = mx + c$
 assume $m = 1$ for simplicity
 $y = x + c$

$$H_t = W_t + c$$

where $c = y$ -intercept of line

predicted = $w_t + \text{intercept}$

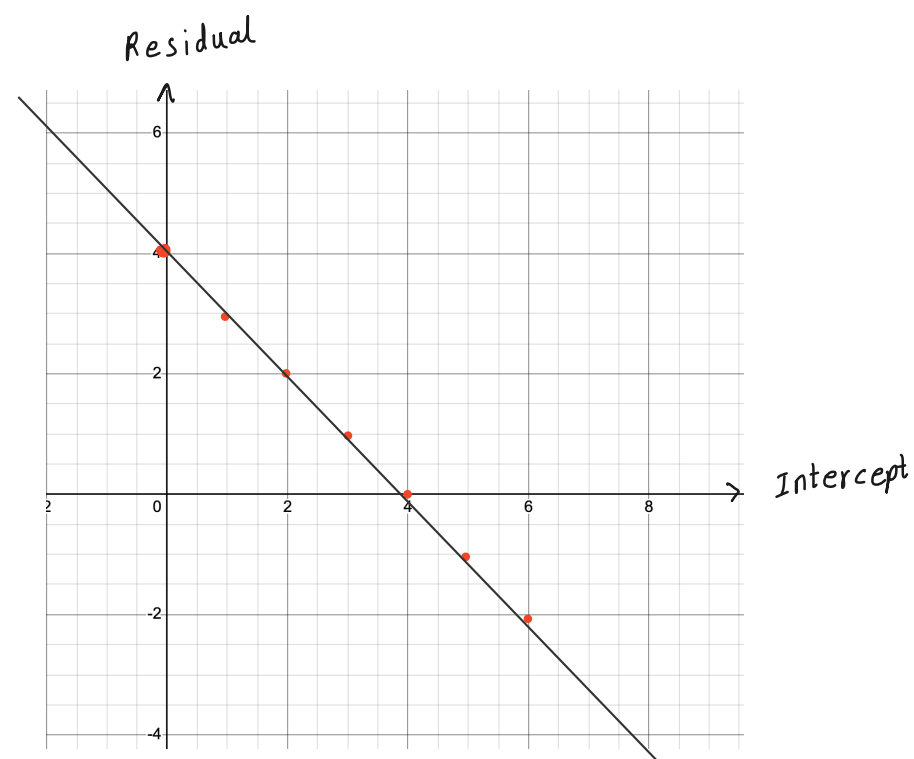
for $W_t = 2$ & $c = 0$

$$\text{Predicted} = 2 + 0 = 2$$

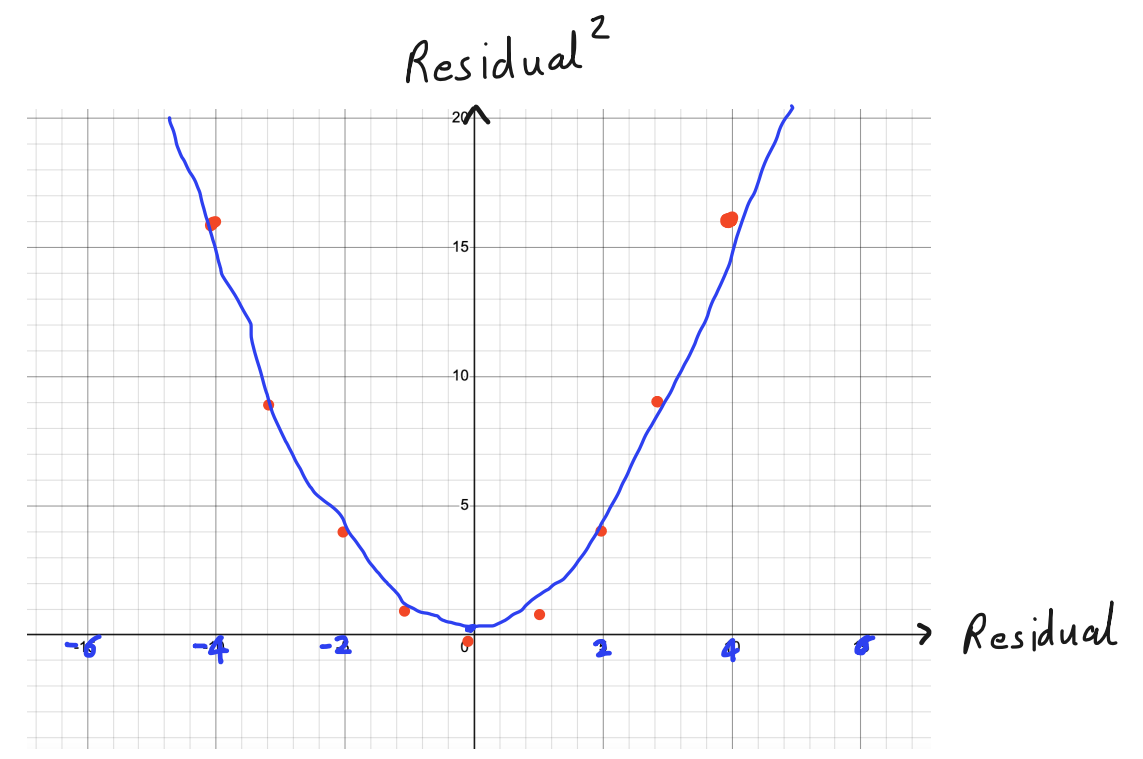
$$\begin{aligned} \text{Residual} &= \text{observed} - \text{predicted} \\ &= 6 - 2 = \underline{\underline{4}} \end{aligned}$$

Plot this on residual vs intercept

Residual Vs Intercept plot



$$\begin{aligned} \frac{d \text{Residual}}{d \text{Intercept}} &= \frac{d}{d \text{Intercept}} (\text{observed} - \text{predicted}) \\ &= \frac{d}{d \text{Intercept}} (\text{observed} - (W_t + \text{Intercept})) \\ &= (0 - (0 + 1)) \\ &= -1 \quad \text{--- (1)} \end{aligned}$$



$$\begin{aligned} \frac{dy}{dx} &= \frac{d \text{Residual}^2}{d \text{Residual}} = \frac{d}{d \text{Residual}} (\text{Residual})^2 \\ &= 2 (\text{Residual}) \quad \text{--- (2)} \end{aligned}$$

To calculate $\frac{d \text{Residual}^2}{d \text{Intercept}}$

using chain Rule

$$\frac{d \text{Residual}^2}{d \text{Intercept}} = \frac{d \text{Residual}^2}{d \text{Residual}} \times \frac{d \text{Residual}}{d \text{Intercept}}$$

using eq (1) & eq (2)

$$\begin{aligned} \frac{d \text{Residual}^2}{d \text{Intercept}} &= 2 (\text{Residual}) \times (-1) \\ &= -2 (\text{Residual}) \\ &= -2 (\text{Observed} - \text{Predicted}) \\ &= -2 (\text{Observed} - (W_t + \text{intercept})) \end{aligned}$$

To optimized residual $\frac{d \text{Residual}^2}{d \text{Intercept}} = 0$

$$\begin{aligned} \therefore 0 &= -2 (\text{Observed} - (W_t + \text{intercept})) \\ 0 &= -2 (6 - (2 + \text{intercept})) \\ 0 &= 6 - 2 - \text{intercept} \\ \therefore \text{intercept} &= 4 \end{aligned}$$